

Advanced Scan Types Execute and compare different scan types against the target Windows VM (192.168.56.105).

Scan type comparison table

```
(kali@kali)-[~]
$ sudo nmap -sS 192.168.56.101
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 05:46 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0037s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
Nmap done: 1 IP address (1 host up) scanned in 5.38 seconds

(kali@kali)-[~]
$ nmap -sT 192.168.56.105
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 05:48 -0400
Nmap scan report for 192.168.56.105
Host is up (0.0028s latency).
All 1000 scanned ports on 192.168.56.105 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 21.84 seconds

(kali@kali)-[~]
$ nmap -sT 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 05:49 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0033s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
Nmap done: 1 IP address (1 host up) scanned in 5.60 seconds
```

```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 05:52 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00014s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Nmap done: 1 IP address (1 host up) scanned in 0.71 seconds

(kali@kali)-[~]
$ sudo nmap -sN 192.168.56.101~
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 05:54 -0400
Failed to resolve "192.168.56.101~".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.07 seconds

(kali@kali)-[~]
$ sudo nmap -sN 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 05:55 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00030s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Nmap done: 1 IP address (1 host up) scanned in 0.69 seconds

(kali@kali)-[~]
$ sudo nmap -sX 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 05:55 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00012s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Nmap done: 1 IP address (1 host up) scanned in 0.70 seconds

(kali@kali)-[~]
$
```

Scan Type	Flag	Stealth Level	Speed	Detection Risk	Result on 192.168.56.101	Observation / Best For
SYN Scan	-sS	High	Fast	Low	Ports 135, 139, 445 open	Stealthy half-open scan; successfully identified Windows services
TCP Connect Scan	-sT	None	Slower	High	Ports 135, 139, 445 open	Full TCP handshake; useful when sudo/root privileges are unavailable
FIN Scan	-sF	Medium	Fast	Medium	1000 closed TCP ports (reset)	Windows responded with RST packets; ineffective for stealth scanning
NULL Scan	-sN	Medium	Fast	Medium	1000 closed TCP ports (reset)	No TCP flags set; Windows reset packets instead of showing open filtered
XMAS Scan	-sX	Medium	Fast	Medium	1000 closed TCP ports (reset)	FIN + PSH + URG flags used; Windows ignored RFC-style behavior

UDP Scanning UDP scanning is slower and less reliable but critical for finding services like DNS, SNMP, and DHCP

UDP scan results screenshot/log

```
(kali@kali)-[~]
$ sudo nmap -sU -p 53,123,161,137,138 192.168.56.101
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 06:14 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0028s latency).

PORT      STATE      SERVICE
53/udp    open|filtered domain
123/udp   open|filtered ntp
137/udp   open       netbios-ns
138/udp   open|filtered netbios-dgm
161/udp   open|filtered snmp

Nmap done: 1 IP address (1 host up) scanned in 1.99 seconds

(kali@kali)-[~]
$ sudo nmap -sU --top-ports 100 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 06:14 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00064s latency).
Not shown: 99 open|filtered udp ports (no-response)
PORT      STATE      SERVICE
137/udp   open       netbios-ns

Nmap done: 1 IP address (1 host up) scanned in 2.57 seconds

(kali@kali)-[~]
```

UDP Scan Results – Windows VM (192.168.56.101)

Port	Protocol	State	Service	Security Implication
53	UDP	open filtered	DNS (domain)	Could indicate DNS service or firewall filtering; may allow DNS enumeration
123	UDP	open filtered	NTP	Can be abused for time synchronization attacks or amplification
137	UDP	open	NetBIOS Name Service	Can expose hostname, shares, and Windows network information
138	UDP	open filtered	NetBIOS Datagram Service	Used for Windows file/printer sharing communication
161	UDP	open filtered	SNMP	May expose system/network information if enabled

Top 100 UDP Scan Results

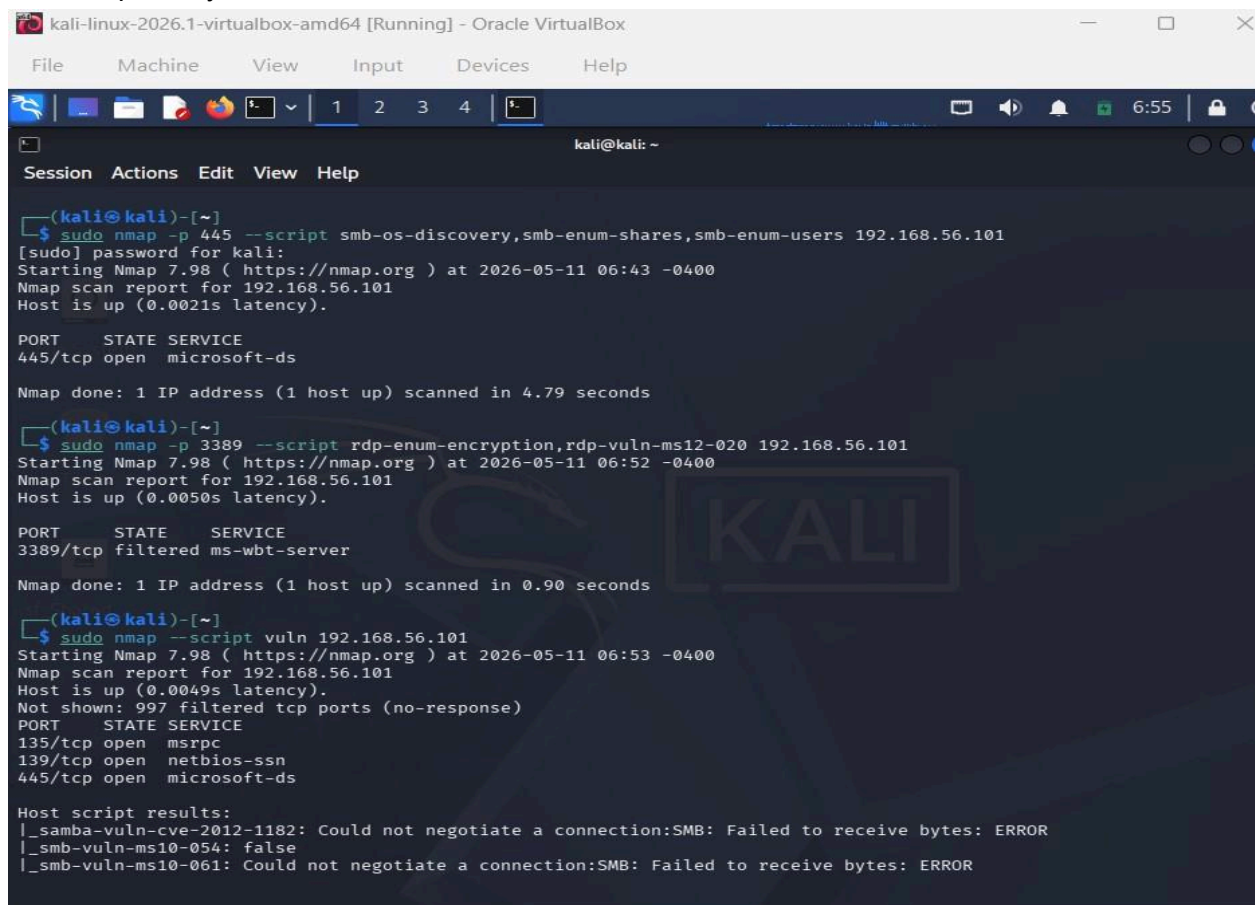
Scan Type Result

sudo nmap -sU --top-ports 100 192.168.56.101 137/udp open netbios-ns

Other 99 Ports open|filtered (no-response)

Nmap Scripting Engine (NSE) NSE transforms Nmap into a vulnerability assessment tool.

NSE script analysis



```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ sudo nmap -p 445 --script smb-os-discovery,smb-enum-shares,smb-enum-users 192.168.56.101
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 06:43 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0021s latency).

PORT      STATE SERVICE
445/tcp    open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 4.79 seconds

(kali@kali)-[~]
$ sudo nmap -p 3389 --script rdp-enum-encryption,rdp-vuln-ms12-020 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 06:52 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0050s latency).

PORT      STATE SERVICE
3389/tcp   filtered ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 0.90 seconds

(kali@kali)-[~]
$ sudo nmap --script vuln 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 06:53 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0049s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

Host script results:
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_smb-vuln-ms10-054: false
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
```

[illegible]

Verification Item	Result
Security Event ID	5152
Meaning	Windows Filtering Platform blocked a packet
Source IP	192.168.56.102 (Kali VM)
Target Ports	135, 139, 445
Detection Outcome	Evasion activity detected and logged

Output Management Professionals save results in all formats using -oA .

Output files (.nmap, .xml, .gnmap)

```
11967,12000,12174,12265,12345,13456,13722,13782-13783,14000,14238,14441-14442,15000,15002-15004,15660,15742,16000-1
6001,16012,16016,16018,16080,16113,16992-16993,17877,17988,18040,18101,18988,19101,19283,19315,19350,19780,19801,19
842,20000,20005,20031,20221-20222,20828,21571,22939,23502,24444,24800,25734-25735,26214,27000,27352-27353,27355-273
56,27715,28201,30000,30718,30951,31038,31337,32768-32785,33354,33899,34571-34573,35500,38292,40193,40911,41511,4251
0,44176,44442-44443,44501,45100,48080,49152-49161,49163,49165,49167,49175-49176,49400,49999-50003,50006,50300,50389
,50500,50636,50800,51103,51493,52673,52822,52848,52869,54045,54328,55055-55056,55555,55600,56737-56738,57294,57797,
58080,60020,60443,61532,61900,62078,63331,64623,64680,65000,65129,65389" />
</extraports>
<port protocol="tcp" portid="135"><state state="open" reason="syn-ack" reason_ttl="64"/><service name="msrpc" produ
ct="Microsoft Windows RPC" ostype="Windows" method="probed" conf="10"><cpe>cpe:/o:microsoft:windows/</cpe></service>
</port>
<port protocol="tcp" portid="139"><state state="open" reason="syn-ack" reason_ttl="64"/><service name="netbios-ssn"
product="Microsoft Windows netbios-ssn" ostype="Windows" method="probed" conf="10"><cpe>cpe:/o:microsoft:windows/<
cpe></service></port>
<port protocol="tcp" portid="445"><state state="open" reason="syn-ack" reason_ttl="64"/><service name="microsoft-ds
" method="table" conf="3"/></port>
</ports>
<times srth="1856" rtvar="2306" to="100000"/>
</host>
<runstats><finished time="1778500106" timestr="Mon May 11 07:48:26 2026" summary="Nmap done at Mon May 11 07:48:26
2026; 1 IP address (1 host up) scanned in 11.95 seconds" elapsed="11.95" exit="success"/><hosts up="1" down="0" tot
al="1"/>
</runstats>
</nmaprun>

(kali@kali)-[~]
$ cat scan_results.gnmap
# Nmap 7.98 Scan initiated Mon May 11 07:48:14 2026 as: /usr/lib/nmap/nmap -sV -oA scan_results 192.168.56.101
Host: 192.168.56.101 () Status: Up
Ports: 135/open/tcp//msrpc//Microsoft Windows RPC/, 139/open/tcp//netbios-ssn//Microsoft Wi
ndows netbios-ssn/, 445/open/tcp//microsoft-ds?/// Ignored State: filtered (997)
# Nmap done at Mon May 11 07:48:26 2026 -- 1 IP address (1 host up) scanned in 11.95 seconds

(kali@kali)-[~]
$ grep open scan_results.gnmap
Host: 192.168.56.101 () Ports: 135/open/tcp//msrpc//Microsoft Windows RPC/, 139/open/tcp//netbios-ssn//Microsoft Wi
ndows netbios-ssn/, 445/open/tcp//microsoft-ds?/// Ignored State: filtered (997)

(kali@kali)-[~]
$
```

NMAP AUTOMATOR SCRIPT The nmap_automator.sh script standardizes the reconnaissance process.

Automated script performance log

```
-(kali@kali)-[~]
$ cat scan_results.xml
?xml version="1.0" encoding="UTF-8"?
!DOCTYPE nmaprun
?xml:stylesheet href="file:///usr/share/nmap/nmap.xsl" type="text/xsl"?
Nmap 7.98 scan initiated Mon May 11 07:48:14 2026 as: /usr/lib/nmap/nmap -sV -oA scan_results 192.168.56.101
nmaprun scanner="nmap" args="/usr/lib/nmap/nmap -sV -oA scan_results 192.168.56.101" start="1778500094" startstr="
on May 11 07:48:14 2026" version="7.98" xmloutputversion="1.05">
scaninfo type="syn" protocol="tcp" numservices="1000" services="1,3-4,6-7,9,13,17,19-26,30,32-33,37,42-43,49,53,70
79-85,88-90,99-100,106,109-111,113,119,125,135,139,143-144,146,161,163,179,199,211-212,222,254-256,259,264,280,301
306,311,340,366,389,406-407,416-417,425,427,443-445,458,464-465,481,497,500,512-515,524,541,543-545,548,554-555,56
587,593,616-617,625,631,636,646,648,666-668,683,687,691,700,705,711,714,720,722,726,749,765,777,783,787,800-801,8
8,843,873,880,888,898,900-903,911-912,981,987,990,992-993,995,999-1002,1007,1009-1011,1021-1100,1102,1104-1108,111
1114,1117,1119,1121-1124,1126,1130-1132,1137-1138,1141,1145,1147-1149,1151-1152,1154,1163-1166,1169,1174-1175,118
1185-1187,1192,1198-1199,1201,1213,1216-1218,1233-1234,1236,1244,1247-1248,1259,1271-1272,1277,1287,1296,1300-130
1309-1311,1322,1328,1334,1352,1417,1433-1434,1443,1455,1461,1494,1500-1501,1503,1521,1524,1533,1556,1580,1583,159
1600,1641,1658,1666,1687-1688,1700,1717-1721,1723,1755,1761,1782-1783,1801,1805,1812,1839-1840,1862-1864,1875,190
1914,1935,1947,1971-1972,1974,1984,1998-2010,2013,2020-2022,2030,2033-2035,2038,2040-2043,2045-2049,2065,2068,209
2100,2103,2105-2107,2111,2119,2121,2126,2135,2144,2160-2161,2170,2179,2190-2191,2196,2200,2222,2251,2260,2268,230
2323,2366,2381-2383,2393-2394,2399,2401,2492,2500,2522,2525,2557,2601-2602,2604-2605,2607-2608,2638,2701-2702,271
2717-2718,2725,2800,2809,2811,2869,2875,2909-2910,2920,2967-2968,2998,3000-3001,3003,3005-3006,3011,3017,3030-303
3052,3071,3077,3128,3168,3211,3221,3260-3261,3268-3269,3283,3300-3301,3306,3322-3325,3333,3351,3367,3369-3372,338
3390,3404,3476,3493,3517,3527,3546,3551,3580,3659,3689-3690,3703,3727,3766,3784,3800-3801,3809,3814,3826-3828,385
3869,3871,3878,3880,3889,3905,3914,3918,3920,3945,3971,3986,3995,3998,4000-4006,4045,4111,4125-4126,4129,4224,424
4279,4321,4343,4443-4446,4449,4550,4567,4662,4848,4899-4900,4998,5000-5004,5009,5030,5033,5050-5051,5054,5060-506
5080,5087,5100-5102,5120,5190,5200,5214,5221-5222,5225-5226,5269,5280,5298,5357,5405,5414,5431-5432,5440,5500,551
5544,5550,5555,5560,5631,5633,5666,5678-5679,5718,5730,5800-5802,5810-5811,5815,5822,5825,5850,5859,5862,587
5900-5906,5907,5910-5911,5915,5922,5925,5950,5952,5959-5963,5985-5989,5998-6007,6009,6025,6059,6100-6101,610
6112,6123,6129,6156,6346,6389,6502,6510,6543,6547,6565-6567,6580,6646,6666-6669,6689,6692,6699,6779,6788-6789,679
6839,6881,6901,6969,7000-7002,7004,7007,7019,7025,7070,7100,7103,7106,7200-7201,7402,7435,7443,7496,7512,7625,762
7676,7741,7777-7778,7800,7911,7920-7921,7937-7938,7999-8002,8007-8011,8021-8022,8031,8042,8045,8080-8090,8093,809
8100,8180-8181,8192-8194,8200,8222,8254,8290-8292,8300,8333,8383,8400,8402,8443,8500,8600,8649,8651-8652,8654,870
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9910,9911,9912,9913,9914,9915,9916,9917,9918,9919,9920,9921,9922,9923,9924,9925,9926,9927,9928,9929,9930,9931,9932,9933,9934,9935,9936,9937,9938,9939,9940,9941,9942,9943,9944,9945,9946,9947,9948,9949,9950,9951,9952,9953,9954,9955,9956,9957,9958,9959,9960,9961,9962,9963,9964,9965,9966,9967,9968,9969,9970,9971,9972,9973,9974,9975,9976,9977,9978,9979,9980,9981,9982,9983,9984,9985,9986,9987,9988,9989,9990,9991,9992,9993,9994,9995,9996,9997,9998,9999,10000" />
</nmaprun>
```

```

Session Actions Edit View Help
└─$ chmod +x nmap_automator.sh

(kali@kali)~$
└─$ ./nmap_automator.sh 192.168.56.101
[+] Starting reconnaissance on 192.168.56.101
[+] Output will be saved to: GraySentinel_Commissioning/1_Portfolio/Reports/nmap_scan_192.168.56.101_20260511_075820
[+] Running Host Discovery...
[+] Running SYN Scan...
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 07:58 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0030s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 5.10 seconds
[+] Running Service Detection...
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 07:58 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0036s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.25 seconds
[+] Running UDP Scan...
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 07:58 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0033s latency).

PORT      STATE SERVICE
53/udp     open/filtered domain
123/udp     open/filtered ntp
137/udp     open      netbios-ns
138/udp     open/filtered netbios-dgm
161/udp     open/filtered snmp

Nmap done: 1 IP address (1 host up) scanned in 2.08 seconds
[+] Running SMB Enumeration...
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-11 07:58 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0087s latency).

PORT      STATE SERVICE

```

```

GNU nano 8.7.1
#!/bin/bash

# nmap_automator.sh - Automated Nmap Recon Script

TARGET=$1

if [ -z "$TARGET" ]; then
    echo "Usage: ./nmap_automator.sh <IP>"
    exit 1
fi

OUTPUT_DIR="GraySentinel_Commissioning/1_Portfolio/Reports"
mkdir -p "$OUTPUT_DIR"

TIMESTAMP=$(date +"%Y%m%d_%H%M%S")
OUTPUT_PREFIX="${OUTPUT_DIR}/nmap_scan_${TARGET}_${TIMESTAMP}"

echo "[+] Starting reconnaissance on $TARGET"
echo "[+] Output will be saved to: $OUTPUT_PREFIX"

# Host Discovery
echo "[+] Running Host Discovery..."
nmap -sn "$TARGET" > "${OUTPUT_PREFIX}_host_discovery.txt"

# TCP SYN Scan
echo "[+] Running SYN Scan..."
sudo nmap -sS "$TARGET" -oA "${OUTPUT_PREFIX}_syn_scan"

# Service Detection
echo "[+] Running Service Detection..."
sudo nmap -sV "$TARGET" -oA "${OUTPUT_PREFIX}_service_scan"

# UDP Scan
echo "[+] Running UDP Scan..."
sudo nmap -sU -p 53,123,137,138,161 "$TARGET" -oA "${OUTPUT_PREFIX}_udp_scan"

# NSE SMB Enumeration
echo "[+] Running SMB Enumeration..."
sudo nmap -p 445 --script smb-os-discovery,smb-enum-shares,smb-enum-users "$TARGET" \
-oA "${OUTPUT_PREFIX}_smb_enum"

echo "[+] Scan completed!"
echo "[+] Results saved in: $OUTPUT_DIR"

```

Successfully executed the Nmap Automator script to automate network reconnaissance on the Windows VM. Performed host discovery, SYN scanning, service/version detection, UDP scanning, and SMB enumeration. Identified open services on ports 135 (MSRPC), 139 (NetBIOS), and 445 (SMB), with 137/udp detected as open for NetBIOS Name Service. Generated structured scan outputs in multiple formats for documentation and analysis.